

Education

Harvard University

Anticipated PhD in Statistics, advised by Prof. Marinka Zitnik

Cambridge, MA

Aug. 2023–present

University of Oxford

MSc in Statistics, advised by Prof. Yee Whye Teh

Oxford, UK

Oct. 2021–June 2023

Stanford University

B.S. in Computer Science and B.A. in Sociology with Honors – GPA: 3.9

Stanford, CA

Sept. 2017–June 2021

Honors & Awards

2023	James Mill Peirce Fellowship , Harvard University	Cambridge, MA
2023	Graduate Research Fellowship , National Science Foundation	Cambridge, MA
2021	Rhodes Scholarship , Rhodes Trust	Oxford, UK
2021	Outstanding Senior Thesis Award and Firestone Medal , Stanford University	Stanford, CA
2020	Spotlight Talk , Computational Biology Workshop, ICML	Vienna, AT
2017	G.R.E.A.T. Award , NHGRI, National Institutes of Health	Bethesda, MD

Publications

Published and accepted

Nic Fishman, Gokul Gowri, Tanush Kumar, Jiaqi Lu, Valentin De Bortoli, Jonathan S. Gootenberg, and Omar Abudayyeh. **Count Bridges enable Modeling and Deconvolving Transcriptomic Data**. *International Conference on Learning Representations*, 2026.

Jonathan Ben-Menachem, Ari Galper, and **Nic Fishman**. **Rethinking Repeatability in Observational Social Science**. *Sociological Methods & Research (accepted)*, 2026.

Nic Fishman, Gokul Gowri, Peng Yin, Jonathan Gootenberg, and Omar Abudayyeh. **Generative Distribution Embeddings**. *Advances in Neural Information Processing Systems*, 2025.

Alexander C Furnas, **Nic Fishman**, Leah Rosenstiel, and Dashun Wang. **Partisan disparities in the funding of science in the United States**. *Science*, 2025.

Jake Fawkes, **Nic Fishman**, Mel Andrews, and Zachary C Lipton. **The fragility of fairness: Causal sensitivity analysis for fair machine learning**. *Advances in Neural Information Processing Systems*, 2024.

Hamed Nilforoshan, Wenli Looi, Emma Pierson, Blanca Villanueva, **Nic Fishman**, Yiling Chen, John Sholar, Beth Redbird, David Grusky, and Jure Leskovec. **Human mobility networks reveal increased segregation in large cities**. *Nature*, 2023.

Nic Fishman, Leo Klarner, Valentin De Bortoli, Emile Mathieu, and Michael Hutchinson. **Diffusion models for constrained domains**. *Transactions on Machine Learning Research*, 2023.

Nic Fishman, Leo Klarner, Emile Mathieu, Michael Hutchinson, and Valentin De Bortoli. **Metropolis sampling for constrained diffusion models**. *Advances in Neural Information Processing Systems*, 2023.

Nic Fishman and Nicholas T Davis. **Change we can believe in: Structural and content dynamics within belief networks**. *American Journal of Political Science*, 2022.

Nic Fishman and Leif Hancox-Li. **Should attention be all we need? The epistemic and ethical implications of unification in machine learning.** *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, 2022.

Nic Fishman, Avanti Shrikumar, Georgi Marinov, and Anshul Kundaje. **Systematic characterization of generative models for de novo design of regulatory DNA.** *ICML Computational Biology Workshop (Spotlight)*, 2020.

Preprints and working papers

Nic Fishman, Gokul Gowri, Paolo LB Fischer, Marinka Zitnik, Omar Abudayyeh, and Jonathan Gootenberg. **Distribution-Conditioned Transport.** *arXiv preprint arXiv:2603.04736*, 2026.

Nic Fishman. **Making (Global) Criminal Procedure: Empire, the End of Justice, and the Rise of Efficiency.** *Stanford University*, 2021.

Presentations

2025	Generative modeling on the space of empirical measures , Boston Diffusion Day	Cambridge, MA
2025	Generative modeling on the space of empirical measures , Harvard Statistics Seminar	Cambridge, MA
2025	False Promises and False Premises of Fair Machine Learning , Columbia Algorithms and Society Seminar	New York, NY
2025	Predicting perturbations at single-cell resolution , Cell Circuits and Epigenomics Seminar	Cambridge, MA
2023	Constrained Riemannian Diffusion Modeling , Thesis Defense	Oxford, UK
2022	Invited talk, Should attention be all we need? , Stanford Human Centered Artificial Intelligence	Stanford, CA
2022	Invited talk, Should attention be all we need? , DeepMind Sociotechnical Systems	London, England
2022	Making (Global) Criminal Procedure , Law and Society Association Annual Meeting	Lisbon, Portugal
2021	Structural and content dynamics within belief networks , American Political Science Association Annual Meeting	Seattle, WA

Research Experience

Harvard Medical School, Harvard University

PhD research with Prof. Marinka Zitnik

Cambridge, MA

Apr. 2024–present

- Building agentic systems to automate statistical method development, with applications in biology and health-care.
- Developing multiscale models for biological systems spanning genomics, single-cell RNA-seq, and spatial transcriptomics, with applications to perturbation prediction.
- Developing EHR-based methods for treatment-effect estimation and policy optimization at health-system scale, including ongoing work with Epic on clinically interpretable decision support.

Harvard Medical School, Harvard University

Research with Profs. Jonathan Gootenberg and Omar Abudayyeh

Cambridge, MA

May 2024–present

- Developing transport and deconvolution methods for single-cell perturbation modeling, enabling generalization across unseen perturbations and cellular contexts.
- Designing transformer architectures for set-valued representations of cell populations to support fine-grained *in silico* experimentation and response prediction.
- Led benchmarking and validation efforts for out-of-distribution prediction in single-cell sequencing, unifying existing methods and evaluation metrics.

Institute for Quantitative Social Science, Harvard University

Research with Prof. Kosuke Imai

Cambridge, MA

June 2020–Sept. 2024

- Constructed a non-parametric framework for causal inference under general interference using regression.
- Developed efficient algorithms and theory for non-parametric monotone regression for estimation and inference in the general interference framework, with applications in large-scale network and spatial experiments.
- Worked on adaptive methods for pure exploration in combinatorial designs in generalized linear models.

Statistics Department, University of Oxford

Research in OxCSML Group under Yee Whye Teh

Oxford, England

Sept. 2021–June 2023

- Built constrained diffusion and flow-matching models for cyclic peptides and antibody loops, connecting generative modeling to molecular design problems.
- Developed techniques for extending diffusion models to constrained domains in general Riemannian geometries.

Center for Science of Science & Innovation, Northwestern University

Research Fellow

Evanston, IL

June 2021–Sept. 2023

- Synthesized huge international datasets on scientific grantmaking and publication to generate a picture of the landscape and political implications of scientific funding.
- Leveraged LLMs to develop fine-grained topic classifications across grants and paper abstracts to understand when grant-making institutions lead scientific investigation as opposed to following what is already being studied.

Economics Department, Massachusetts Institute of Technology

Research Assistant for Prof. Victor Chernozhukov

Cambridge, MA

June 2021–June 2022

- Developed methods for dynamic treatment effect estimation and policy learning in healthcare.
- Specifically extended g-estimation to non-linear blip functions using tools from minimax optimization.

Stanford Center on Poverty and Inequality and Stanford Network Analysis Project

Research Intern for Prof. Jure Leskovec and Prof. David Grusky

Stanford, CA

Jan. 2020–June 2021

- Used path crossings to study socioeconomic stratification in the United States, developing insights about heterogeneity in segregation across different social contexts and across different income deciles.
- Worked on algorithms to identify when individuals cross paths in time and space using massive GPS data.

Kundaje Lab, Stanford AI Lab

Research Intern for Prof. Anshul Kundaje

Stanford, CA

Jan. 2018–June 2021

- Created modular system for generating DNA sequences using GANs/VAEs/Transformers and optimizing these models to produce sequences with specified properties (level of gene expression or chromatin accessibility).
- Developed a k-NN algorithm to evaluate (1) the fidelity of samples from generative models and (2) the robustness of neural network predictions on regression outputs (extending existing work on classification).

Work Experience

Knowledge Graph Team, Google

Research Intern

San Francisco, CA

summer 2022

- Research integrating LLMs and knowledge graphs (KGs) for search, particularly combining LLM representations with graph neural networks over a knowledge graph to produce KG-informed representations.
- Developed methods to identify author expertise using LLM/KG representations for multimedia topic modeling.

Data for Progress

Data Scientist

New York, NY

June 2018–May 2021

- Led development of polling infrastructure that produced the most accurate poll results in the Democratic Primary. Built out MySQL database for storage and analysis of survey responses. Automated chart and report generation from this database. Developed search engine and website for internal use to assist in research.
- Designed, conducted, and analyzed polls used to guide policy change for the Green New Deal, Medicare for All, several HR-1 issues, and criminal justice reform, among other progressive issues.
- Developed novel poll weighting scheme achieving state-of-the-art accuracy.
- Automated argument detection so that non-technical colleagues to easily interpret open-ended survey responses in policy briefs by building hierarchical Dirichlet process models for non-parametric topic modeling.
- Created an ecological loss function and corresponding neural networks, extending ecological inference.
- Led team creating word2vec models to analyze gender/racial bias in news articles around the 2016 election.

Sunrise Movement

Data Science Consultant

Washington, DC

fall 2020

- Used LASSO and random forest algorithms to build interpretable heuristics for identifying low-propensity swing state voters for a large-scale get-out-to-vote campaign in advance of the 2020 presidential election.

Star Lab Corporation

Machine Learning Specialist

Washington, DC

summer 2018

- Worked on a Red Hat kernel module to log system activity and leveraged it for neural net anomaly detection.